Fitness For Purpose: (FFP)

User Manual

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Version: 3.2.x

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Introduction

1.1 Overview

The **Pipe Tally Comparison Tool** is a desktop-based engineering application designed to compare two pipeline tally sheets side-by-side and determine the similarity, differences, and consistency of pipeline features recorded during In-Line Inspection (ILI) or field measurement activities. The software automates the feature-by-feature comparison process using predefined engineering tolerances and presents results in an intuitive, color-coded interface suitable for technical analysis.

Pipeline integrity assessments often involve validating multiple inspection datasets—such as comparing a new ILI run with a previous run or validating construction tally sheets against as-built records. Manual comparison of these datasets is time-consuming, prone to human error, and difficult to standardize. This tool solves that challenge by performing automated matching, corrosion feature comparison, weld alignment identification, and generating structured outputs for documentation.

1.2 What is FFP (Fitness-For-Purpose)?

The FFP is an engineering methodology used in pipeline integrity to determine whether a pipeline or a specific feature (corrosion, dent, metal loss, etc.) is fit to continue operating safely under current or proposed conditions.
It focuses on answering three questions:

1. **Is the pipeline safe to operate?**
2. **How severe are the identified defects?**
3. **Is immediate repair required, or can operation continue with monitoring?**

Installation and Setup

2.1 System Requirements

Minimum RAM: 4GB

Minimum Processor: i5 and (or equivalent) 5th generation

Operating System: Windows 10/11

Disk Space: Minimum of 2GB of free disk space for installation

Graphics Card: DirectX 11 compatible graphics card with at least 512MB of VRAM (for optimal performance)

Getting Started

3.1 Data

The **Pipe Tally Comparison Tool** operates on two pipeline tally sheets—**Pipe Tally 1** and **Pipe Tally 2**—which are provided by the user in **Excel (.xlsx)** format. These tallies contain detailed, feature-by-feature records collected during pipeline inspections, construction, or verification activities.

The software reads both tallies, standardizes their column structure, and performs automated comparison based on defined engineering rules.

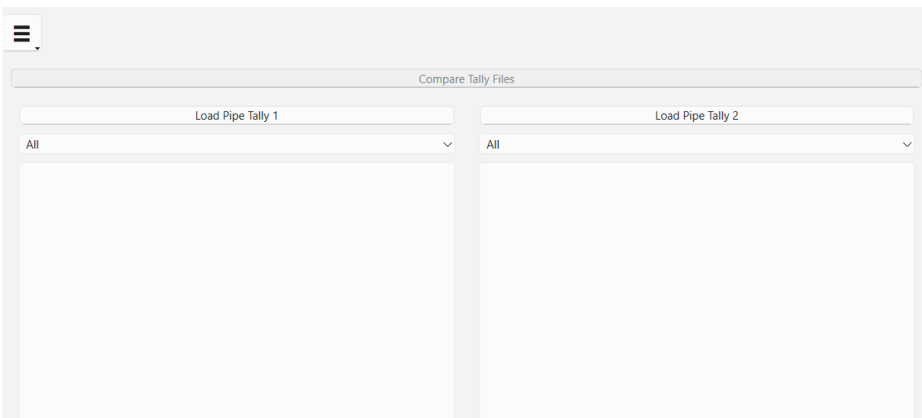
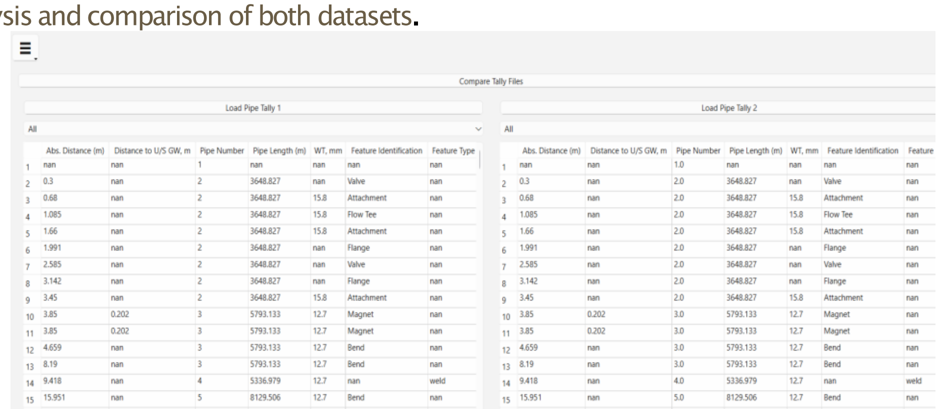


Fig-3.1

3.2 First-Time Use

Once both pipe tally files (Pipe Tally 1 and Pipe Tally 2) are successfully loaded, the "Compare Tally Files" button becomes active. This button triggers the core functionality of the software—automatic analysis and comparison of both datasets.



	Abs. Distance (m)	Distance to U/S GW, m	Pipe Number	Pipe Length (m)	WT, mm	Feature Identification	Feature Type
1	nan	nan	1	nan	nan	nan	nan
2	0.3	nan	2	3648.827	nan	Valve	nan
3	0.68	nan	2	3648.827	15.8	Attachment	nan
4	1.085	nan	2	3648.827	15.8	Flow Tee	nan
5	1.66	nan	2	3648.827	15.8	Attachment	nan
6	1.991	nan	2	3648.827	nan	Flange	nan
7	2.585	nan	2	3648.827	nan	Valve	nan
8	3.142	nan	2	3648.827	nan	Flange	nan
9	3.45	nan	2	3648.827	15.8	Attachment	nan
10	3.85	0.202	3	5793.133	12.7	Magnet	nan
11	3.85	0.202	3	5793.133	12.7	Magnet	nan
12	4.659	nan	3	5793.133	12.7	Bend	nan
13	8.19	nan	3	5793.133	12.7	Bend	nan
14	9.418	nan	4	5336.979	12.7	nan	weld
15	15.951	nan	5	8129.506	12.7	Bend	nan
16	20.535	nan	5	8129.506	nan	Flange	nan

	Abs. Distance (m)	Distance to U/S GW, m	Pipe Number	Pipe Length (m)	WT, mm	Feature Identification	Feature
1	nan	nan	1.0	nan	nan	nan	nan
2	0.3	nan	2.0	3648.827	nan	Valve	nan
3	0.68	nan	2.0	3648.827	15.8	Attachment	nan
4	1.085	nan	2.0	3648.827	15.8	Flow Tee	nan
5	1.66	nan	2.0	3648.827	15.8	Attachment	nan
6	1.991	nan	2.0	3648.827	nan	Flange	nan
7	2.585	nan	2.0	3648.827	nan	Valve	nan
8	3.142	nan	2.0	3648.827	nan	Flange	nan
9	3.45	nan	2.0	3648.827	15.8	Attachment	nan
10	3.85	0.202	3.0	5793.133	12.7	Magnet	nan
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12	4.659	nan	3.0	5793.133	12.7	Bend	nan
13	8.19	nan	3.0	5793.133	12.7	Bend	nan
14	9.418	nan	4.0	5336.979	12.7	nan	weld
15	15.951	nan	5.0	8129.506	12.7	Bend	nan
16	20.535	nan	5.0	8129.506	nan	Flange	nan

Fig-3.2

User Interface

4.1 Comparison Result Screen

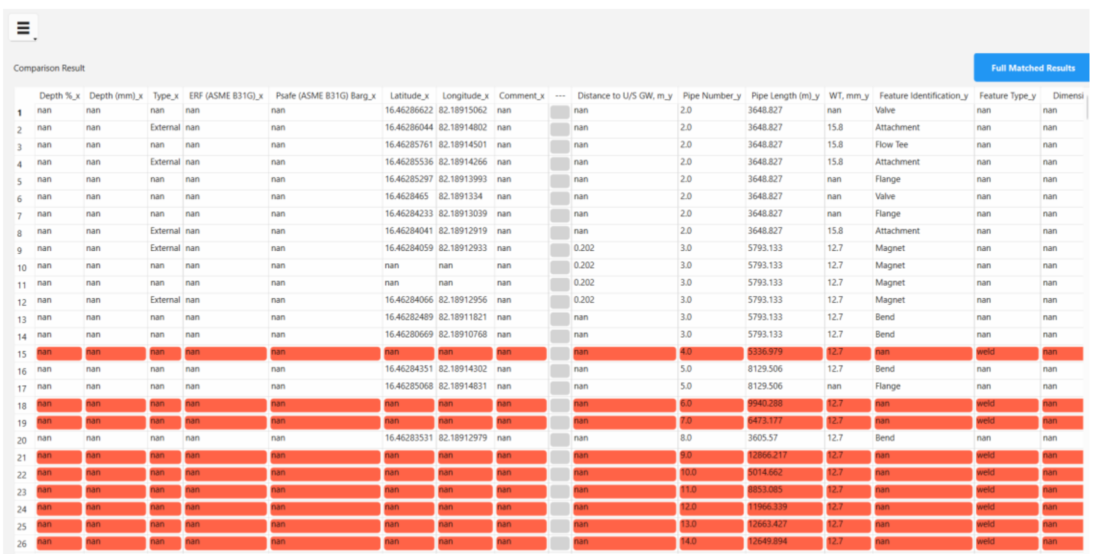
Some After loading both pipe tally files and clicking the Compare Tally Files button, the user is automatically directed to the Comparison Result Screen. This is the primary analysis view where the software displays a combined and color-coded comparison of both tallies.

Key Features of This Screen

1. Side-by-Side Data View

- The left tally appears with column names ending in **_x**.
- The right tally appears with column names ending in **_y**.
- A **grey separator column** (---) visually divides the two datasets.

This makes it clear which values belong to which tally.



Comparison Result

Full Matched Results

	Depth %_x	Depth (mm)_x	Type_x	ERF (ASME B31G)_x	Psafe (ASME B31G) Barg_x	Latitude_x	Longitude_x	Comment_x	---	Distance to U/S GW, m_y	Pipe Number_y	Pipe Length (m)_y	WT, mm_y	Feature Identification_y	Feature Type_y	Dimensi
1	nan	nan	nan	nan	nan	16.46286622	82.18915062	nan		nan	2.0	3648.827	nan	Valve	nan	nan
2	nan	nan	External	nan	nan	16.46286044	82.18914802	nan		nan	2.0	3648.827	15.8	Attachment	nan	nan
3	nan	nan	nan	nan	nan	16.46285761	82.18914501	nan		nan	2.0	3648.827	15.8	Flow Tee	nan	nan
4	nan	nan	External	nan	nan	16.46285536	82.18914266	nan		nan	2.0	3648.827	15.8	Attachment	nan	nan
5	nan	nan	nan	nan	nan	16.46285297	82.18913993	nan		nan	2.0	3648.827	nan	Flange	nan	nan
6	nan	nan	nan	nan	nan	16.4628465	82.1891334	nan		nan	2.0	3648.827	nan	Valve	nan	nan
7	nan	nan	nan	nan	nan	16.46284233	82.18913039	nan		nan	2.0	3648.827	nan	Flange	nan	nan
8	nan	nan	External	nan	nan	16.46284041	82.18912919	nan		nan	2.0	3648.827	15.8	Attachment	nan	nan
9	nan	nan	External	nan	nan	16.46284059	82.18912933	nan		0.202	3.0	5793.133	12.7	Magnet	nan	nan
10	nan	nan	nan	nan	nan	nan	nan	nan		0.202	3.0	5793.133	12.7	Magnet	nan	nan
11	nan	nan	nan	nan	nan	nan	nan	nan		0.202	3.0	5793.133	12.7	Magnet	nan	nan
12	nan	nan	External	nan	nan	16.46284066	82.18912956	nan		0.202	3.0	5793.133	12.7	Magnet	nan	nan
13	nan	nan	nan	nan	nan	16.46282489	82.18911821	nan		nan	3.0	5793.133	12.7	Bend	nan	nan
14	nan	nan	nan	nan	nan	16.46280669	82.18910768	nan		nan	3.0	5793.133	12.7	Bend	nan	nan
15	nan	nan	nan	nan	nan	nan	nan	nan		nan	4.0	5336.979	12.7	nan	weld	nan
16	nan	nan	nan	nan	nan	16.46284351	82.18914302	nan		nan	5.0	8129.506	12.7	Bend	nan	nan
17	nan	nan	nan	nan	nan	16.46285068	82.18914831	nan		nan	5.0	8129.506	nan	Flange	nan	nan
18	nan	nan	nan	nan	nan	nan	nan	nan		nan	6.0	9940.288	12.7	nan	weld	nan
19	nan	nan	nan	nan	nan	nan	nan	nan		nan	7.0	6473.177	12.7	nan	weld	nan
20	nan	nan	nan	nan	nan	16.46283531	82.18912979	nan		nan	8.0	3605.57	12.7	Bend	nan	nan
21	nan	nan	nan	nan	nan	nan	nan	nan		nan	9.0	12866.217	12.7	nan	weld	nan
22	nan	nan	nan	nan	nan	nan	nan	nan		nan	10.0	5014.662	12.7	nan	weld	nan
23	nan	nan	nan	nan	nan	nan	nan	nan		nan	11.0	8853.085	12.7	nan	weld	nan
24	nan	nan	nan	nan	nan	nan	nan	nan		nan	12.0	11966.339	12.7	nan	weld	nan
25	nan	nan	nan	nan	nan	nan	nan	nan		nan	13.0	12663.427	12.7	nan	weld	nan
26	nan	nan	nan	nan	nan	nan	nan	nan		nan	14.0	12649.894	12.7	nan	weld	nan

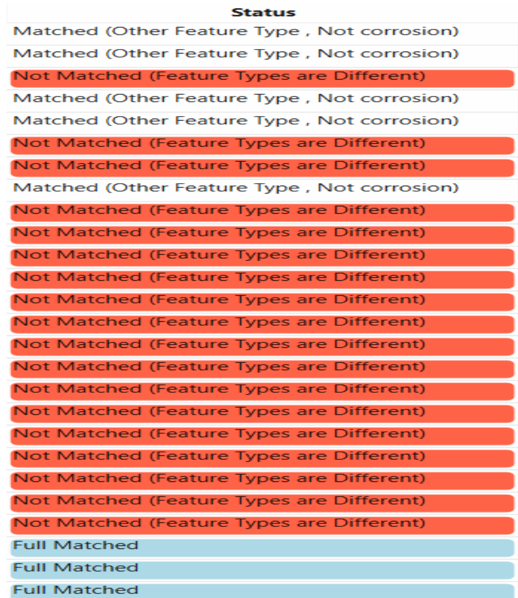
Fig-4.1.1

2. Status Column

Every row includes a final **Status** column that indicates how well the feature matched between the two tallies. Example statuses include:

- **Full Matched**
- **Matched First Weld**
- **Matched (Other Feature)**
- **Not Matched (Corrosion)**
- **Not Matched (Types Different)**
- **Unmatched**

This status is the core output used for engineering assessment.



Status
Matched (Other Feature Type , Not corrosion)
Matched (Other Feature Type , Not corrosion)
Not Matched (Feature Types are Different)
Matched (Other Feature Type , Not corrosion)
Matched (Other Feature Type , Not corrosion)
Not Matched (Feature Types are Different)
Not Matched (Feature Types are Different)
Matched (Other Feature Type , Not corrosion)
Not Matched (Feature Types are Different)
Not Matched (Feature Types are Different)
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Not Matched (Feature Types are Different)
Not Matched (Feature Types are Different)
Not Matched (Feature Types are Different)
Not Matched (Feature Types are Different)
Not Matched (Feature Types are Different)
Full Matched
Full Matched
Full Matched

Fig-4.1.2

3. Color-Coded Results

Cells are automatically highlighted based on match status:

- **Light Blue** → Fully matched
- **White** → Matched, not corrosion
- **Red** / **Tomato** → Not matched or mismatched data

This visual representation helps quickly identify inconsistencies, missing features, and new anomalies.

4.2 Full Matched Results Screen

When the user clicks the Full Matched Results button on the top right of the Comparison Result Screen, the software navigates to the Full Matched Results Screen, shown in the second image.

This screen contains **only those rows that were identified as Full Matched**, making it ideal for documentation and defect tracking.

Key Features of This Screen

1. Filtered Data View

Only features that meet *all* matching criteria (distance, dimensions, orientation, weld spacing, etc.) are displayed.

This provides a clean and focused dataset for:

- Growth rate calculations
- Repeatability studies
- Corrosion verification
- FFP reporting

2. Consistent Side-by-Side Column Layout

Just like the comparison screen:

- Columns ending with **_x** belong to Pipe Tally 1
- Columns ending with **_y** belong to Pipe Tally 2
- A separator (---) splits the two data groups

3. Export Functionality

A **Download** button allows exporting the filtered full-matched data into an Excel file (.xlsx).
The exported file includes:

- Colour coding
- Auto column-width formatting
- Grouped column styling (yellow for _x, blue for _y)

This ensures the exported file is ready to be placed directly into FFP reports without additional manual formatting.

4. Back to Comparison Button

The **Back to Comparison** button allows users to return to the full comparison view and continue reviewing mismatches and alignment points.

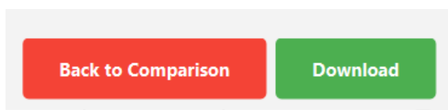

Fig-4.2